

1-Day Technology Update Workshop

AI Disruption and the Future of Cooperative Enterprise Systems

Prepared for: ANGKASA IT Division (Bahagian Teknologi Maklumat)

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Executive Summary

ANGKASA sits at a critical juncture. Its core SPGA salary deduction system — serving **3.5 million cooperative members** across **16,284 cooperatives** — runs on **Unisys ClearPath technology deployed since 1996**. This 30-year-old mainframe backbone has been reliable, but the world around it is changing at unprecedented speed.

This paper presents the case that **any ERP procurement exercise conducted in 2026 without accounting for AI disruption risks locking ANGKASA into another generation of vendor dependency** — precisely at the moment when the economics of enterprise software are being fundamentally rewritten.

The opportunity is equally significant: ANGKASA's unique position as the **apex body for 16,284 cooperatives with RM165.8 billion in assets** makes it an ideal incubator for a **cooperative-owned technology platform** that could serve the entire movement — at a fraction of proprietary costs.

Part 1: The FUD That Should Concern You

1.1 Your ERP RFP May Already Be Outdated

If ANGKASA is evaluating enterprise systems using a traditional RFP framework — comparing SAP, Oracle, and Microsoft on features, pricing, and implementation timelines — the evaluation criteria themselves may be obsolete within 18 months.

Why? Because the ERP market is undergoing its most significant disruption in 35 years:

“Through 2027, GenAI and AI agent use will create the first true challenge to mainstream productivity tools in 35 years, prompting a \$58 billion market shake-up.”
— Gartner, 2025

Key data points your RFP evaluation may not have considered:

Indicator	2024	2026 (Now)	2027 Projection
Enterprise apps with AI agents	<5%	~20%	40% (Gartner)
ERP spending including AI capabilities	14%	~35%	62% (Gartner)
AI-native ERP startups funded	Emerging	DualEntry (\$90M), others	\$270B AI software market
SAP ECC end of support	Active	22 months left	Dec 31, 2027

The risk: Selecting a vendor today based on 2024-era criteria means you are buying a system designed for **the era that is ending**, not the one beginning.

1.2 The SAP 2027 Cliff

SAP has set a hard deadline: **mainstream ECC support ends December 31, 2027.**

Extended maintenance (to 2030) costs a **2% premium** on top of the existing 22% maintenance fee. Organizations that wait face **30-50% higher consulting rates** due to the global migration demand surge.

What this means for ANGKASA: Any vendor proposing SAP is proposing a system whose predecessor is being forcibly retired. The migration track record is sobering:

- **32% of companies** migrating to S/4HANA have **underestimated total cost**

- Typical migration: **12-24 months**, often extending beyond
- Organizations on unsupported ECC face **40-60% higher total cost of ownership**

1.3 Oracle and Microsoft Are Raising Prices

The proprietary ERP vendors are coordinating price increases while bundling AI features that may or may not deliver value:

Vendor	Current Pricing	Recent Changes
Oracle Fusion Cloud ERP	RM33,000/user/year (\$7,500)	3-year minimum contract, minimum 25 users
SAP S/4HANA Cloud	RM31,000-66,000+/user/year	Forced migration from ECC by 2027
Microsoft Dynamics 365	RM9,500-20,600/user/year	First price increase in 5 years (Nov 2025); M365 suite +8-23% effective (Jul 2026)

At Oracle pricing, 100 ANGKASA staff = RM3.3 million/year in license fees alone. At SAP pricing, the same could exceed RM6.6 million/year.

This is before implementation, customisation, training, and ongoing support.

1.4 The AI-Native Disruption Is Real

In October 2025, a startup called **DualEntry** emerged from stealth with **\$90 million** in funding (Lightspeed, Khosla Ventures, Google Ventures) and a provocative claim: they can migrate organisations off legacy ERP systems **in 24 hours**.

Their investor thesis was published under the title: "**The Death of the Legacy ERP**".

Key claims:

- First ERP built from the ground up with AI (not AI bolted onto legacy code)
- Already processed **\$100 billion** in journal entries
- Signed **thousands** of global users in months
- Valued at **\$415 million** in 15 months

Whether DualEntry succeeds is not the point. The point is that **\$100M+ in venture capital is betting that legacy ERP can be replaced by AI-native alternatives**. This is a structural market shift, not a trend.

1.5 Forrester's Warning on Vendor Lock-In

Forrester's 2026 enterprise software predictions contain a specific warning:

"Consolidating to a primary platform concentrates risk and neuters negotiation leverage, making the next major software decision a bet on a single vendor's security posture, pricing model, and innovation capacity for the next decade."

"Vendors are using AI to drive new revenue streams while tightening control... The era of experimentation is over. The era of monetisation has begun."

Translation: The vendors offering AI features are using them as a justification to raise prices and deepen lock-in. An RFP that selects a proprietary vendor today is not just buying software — it is accepting a decade of pricing decisions made by that vendor.

Part 2: The Strategic Position

2.1 ERP Vendor Landscape — Cost vs AI Readiness

The following matrix positions major ERP options on two axes that matter for ANGKASA's decision:

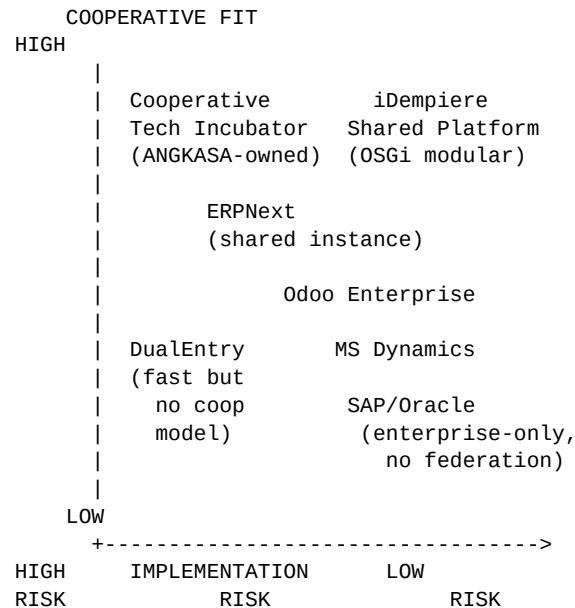
	HIGH COST	LOW COST
HIGH AI READINESS	SAP S/4HANA RM31-66K/user/year Oracle Fusion RM33K/user/year DualEntry (AI-native) (startup, unproven)	OPPORTUNITY ZONE Open Source + AI Layer (iDempiere/ERPNext + local AI integration)
LOW AI READINESS	MS Dynamics RM10-21K/user/year	Odoo Enterprise RM1.3-2K/user/year ERPNext <RM50/user/year iDempiere RM0 license
	COST → decreasing →	

Reading the matrix:

- **Top-left quadrant (High Cost, High AI):** SAP and Oracle are investing heavily in AI, but at extreme cost. ANGKASA would pay RM3-7M/year for 100 users and still be locked into their roadmap.
- **Bottom-right quadrant (Low Cost, Low AI):** Open source ERP (iDempiere, ERPNext) costs almost nothing in licensing but has no native AI today.
- **The Opportunity Zone (center-right):** An open-source ERP base enhanced with a targeted AI layer — built or integrated locally — combines low cost with relevant AI capability. **This is where ANGKASA should position.**

2.2 Deployment Strategy — Implementation Risk vs Cooperative Fit

This matrix evaluates deployment approaches against the unique requirements of a ****federated cooperative movement****:



Reading the matrix:

- **Bottom-left (High Risk, Low Fit):** SAP/Oracle are designed for single enterprises, not federated cooperative movements. Implementation risk is extreme (12-24 months, cost overruns).
- **Top-right (Low Risk, High Fit):** A cooperative-owned technology incubator using open-source ERP scores highest. Proven in the credit union sector (Velera serves 4,000 credit unions; CU*Answers is cooperative-owned core processing).
- **iDempiere** sits in the best position for ANGKASA specifically: Java/OSGi modularity means each cooperative type can have custom plugins, it matches ANGKASA's enterprise IT culture, and it has an active Malaysia community.

2.3 The Cost Reality at ANGKASA's Scale

ANGKASA is not a single company. It is the technology backbone for **16,284 cooperatives** and **7.22 million members**. At this scale, vendor pricing models produce staggering numbers:

Scenario	Annual License Cost (est.)	5-Year TCO (est.)
Oracle Fusion (500 core users)	RM16.5 million	RM82+ million
SAP S/4HANA Cloud (500 core users)	RM15-33 million	RM75-165+ million
Microsoft Dynamics 365 (500 core users)	RM4.8-10.3 million	RM24-52 million
Odoo Enterprise (500 users)	RM650K-1M	RM3.3-5 million
iDempiere (self-hosted, cooperative-owned)	RM0 license	Infrastructure + support only
ERPNext (self-hosted)	<RM25K	Infrastructure + support only

The question is not "Can ANGKASA afford SAP?" The question is: "Can 16,284 cooperatives justify sending RM80-160 million to a foreign vendor over 5 years when cooperative-owned alternatives exist?"

Part 3: ANGKASA as Technology Incubator

3.1 The Precedent: Credit Union Shared Services

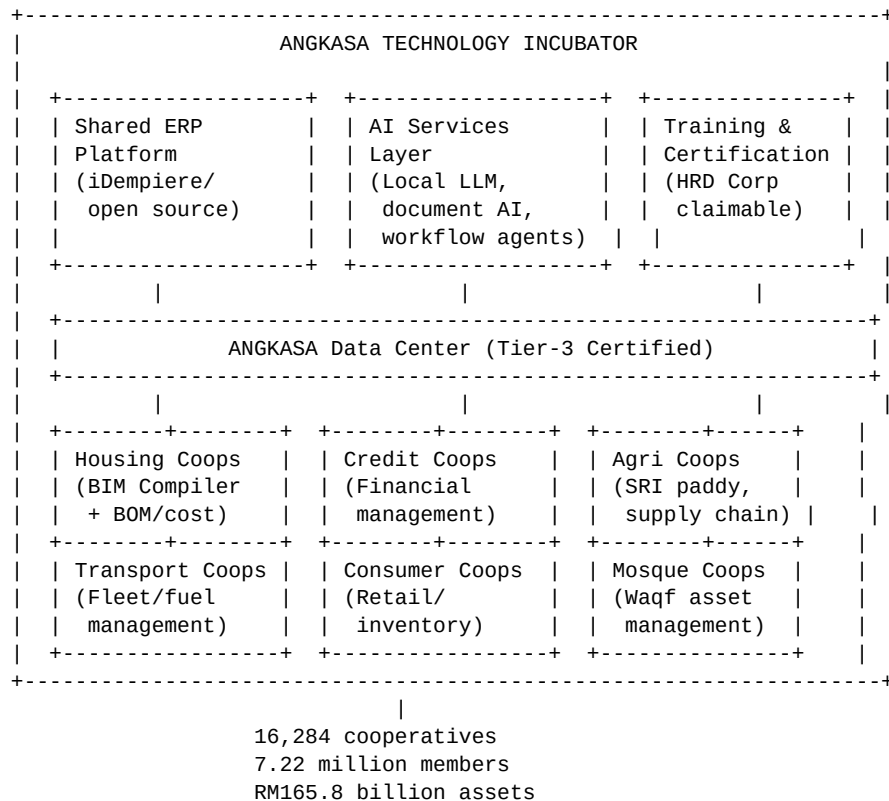
The model already exists — not in theory, but in production at massive scale:

Organisation	Model	Scale
Velera (formerly PSCU/Co-op Solutions)	Cooperative-owned payments CUSO	4,000+ credit unions across North America
CU*Answers (CUaxis/CBX)	Cooperative-owned core data processing	Hundreds of credit unions
Co-op Shared Branch Network	Federated transaction network	5,000+ branches

These are cooperatives building technology platforms for cooperatives. The members own the platform. The costs are shared. The vendor lock-in is zero.

ANGKASA already operates this way for salary deductions (SPGA). The question is: **why not extend the model to ERP?**

3.2 The ANGKASA Incubator Concept



3.3 Why iDempiere Fits ANGKASA

Requirement	iDempiere Capability
Java enterprise stack	Matches ANGKASA's existing IT culture (enterprise-grade, not startup-grade)
OSGi modularity	Each cooperative type gets purpose-built plugins without forking the core
100% open source (GPLv2)	Zero license fees. ANGKASA owns the code. No vendor can raise prices.
Active Malaysia community	Redhuan D. Oon (author of Open Source ERP, Pearson Malaysia) is based in Putrajaya. Local expertise exists.
Manufacturing + Distribution	Modules for supply chain, warehousing, POS — relevant to diverse cooperative types
Multi-organisation architecture	Built-in support for multiple entities sharing one database — ideal for cooperative federation
20+ year track record	Descended from Compiere (2000) →

	ADempiere (2006) → iDempiere (2012). Battle-tested.
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3.4 The AI Layer — Built, Not Bought

Rather than buying AI from SAP or Oracle (at their prices, on their terms), ANGKASA could **build an AI capability layer on top of open-source ERP**:

AI Capability	Approach	Cooperative Use Case
Document processing	Local LLM (Ollama/vLLM) + OCR	Automate invoice/receipt processing for 16K cooperatives
Financial reporting agents	Agentic AI on structured ERP data	Monthly close, audit preparation, regulatory compliance
BIM + Cost estimation	BIM Intent Compiler (open source)	Housing cooperative design → BOM → cost estimate
Credit assessment	AI on SPeKAR + cooperative financial data	Enhance loan assessment for cooperative members
Workflow automation	AI agents replacing manual approval chains	Procurement, HR, membership management

This is the "Opportunity Zone" from the strategy matrix — open source ERP + locally-built AI, owned by the cooperative movement.

Part 4: The DaKoM 2030 Alignment

Malaysia's **Dasar Koperasi Malaysia (DaKoM) 2030** targets:

- Cooperative sector revenue of **RM73 billion** by 2030 (from RM45.5 billion)
- Active cooperative rate of **65%**
- Improvement in Cooperative Member Well-being Index (IndeKA) to **80**

A technology incubator directly supports all three targets:

DaKoM Target	Technology Contribution
RM73B revenue	Shared ERP reduces operational costs, freeing capital for growth. AI automation improves productivity.
65% active rate	Low-cost shared platform makes digital tools accessible to small and inactive cooperatives currently priced out.
IndeKA score 80	Better financial management, faster loan processing, transparent reporting improve member well-being.

Budget 2026 allocated **RM50 million** for cooperative development through SKM. A portion directed toward a cooperative technology incubator would have multiplied impact across all 16,284 cooperatives.

Part 5: Workshop Agenda

Morning — Understanding the Threat

Time	Module	Content
09:00	Opening	Workshop objectives. Why this matters now.
09:15	Module 1: The AI Disruption	Gartner's 40% AI agent prediction. DualEntry and the "Death of Legacy ERP" thesis. What "AI-native" means vs "AI-bolted-on". Demo: AI agent processing a cooperative financial report.
10:15	Break	
10:30	Module 2: The Pricing Trap	SAP 2027 deadline. Oracle's RM33K/user/year. Microsoft's 23% effective increase. Forrester's lock-in warning. Cost projections at ANGKASA's scale. Interactive exercise: calculate your own 5-year TCO under three scenarios.
11:30	Module 3: Your Current Position	ANGKASA's Unisys ClearPath — 30 years of service. SPGA system strengths and risks. Where the gaps are. Facilitated discussion: what keeps the IT team up at night?
12:30	Lunch	

Afternoon — Seeing the Opportunity

Time	Module	Content
13:30	Module 4: The Open Source Alternative	iDempiere, ERPNext, Odoo — live comparison. Total cost of ownership at cooperative scale. Case study: Velera (4,000 credit unions on shared platform).
14:30	Module 5: ANGKASA as Incubator	The cooperative technology incubator model. Architecture walkthrough. How OSGi

		modularity serves 16,284 diverse cooperatives. DaKoM 2030 alignment. Interactive: mapping cooperative types to ERP modules.
15:30	Break	
15:45	Module 6: The AI Layer You Can Own	Local AI deployment (not cloud-dependent). Document processing demo. BIM Compiler demo (housing cooperative use case). Financial reporting agent concept.
16:30	Module 7: Strategic Planning Exercise	9-Quadrant SWOT analysis (Prof. Wan Mohtar's methodology). Teams draft a 90-day action plan. Presentation and critique.
17:15	Closing	Summary of key decisions. Next steps. Commitment to action.

Part 6: Recommended Next Steps for ANGKASA

Priority	Action	Timeline	Owner
1	Pause or reframe any active ERP RFP to include AI-readiness and cooperative-fit criteria	Immediate	IT Division
2	Commission a formal TCO comparison: proprietary vs open-source at ANGKASA scale	30 days	IT Division + Nusa
3	Pilot iDempiere with one cooperative type (e.g., credit cooperative) as proof of concept	90 days	IT Division + Redhuan D. Oon
4	Establish a Technology Incubator steering committee with representation from diverse cooperative types	60 days	Prof. Wan Mohtar + ANGKASA board
5	Present incubator concept to SKM for alignment with DaKoM 2030 and Budget 2026 allocation	Q3 2026	Prof. Wan Mohtar + ANGKASA CEO
6	Develop HRDC-claimable training curriculum for cooperative digital readiness	Q3 2026	Nusa

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